

GREAT DUCK ISLAND, ON, Canada

WMO: 714620

Lat: 45.642N

Lon: 82.962W

Elev: 600

StdP: 14.38

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.7	(c) 1.0	(d) -12.4	(e) 2.9	(f) -3.5	(g) -7.7	(h) 3.7	(i) 1.6	(j) 36.0	(k) 20.7	(l) 32.8	(m) 23.3	(n) 6.9	(o) 0	(p) 0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 9.5	(c) 75.8	(d) 65.5	(e) 74.0	(f) 65.3	(g) 72.4	(h) 64.9	(i) 69.4	(j) 73.8	(k) 68.1	(l) 72.1	(m) 66.8	(n) 70.7	(o) 6.7	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.7	103.9	72.1	66.4	99.5	71.0	65.1	95.0	69.9	33.9	73.9	32.7	72.2	31.7	70.7	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 29.6	(b) 25.8	(c) 22.6	(e) -10.4	(f) 79.7	(g) 6.1	(h) 2.9	(i) -14.8	(j) 81.7	(k) -18.4	(l) 83.4	(m) -21.9	(n) 85.0	(o) -26.3	(p) 87.1		
			DB													
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.4	20.8	21.8	28.1	37.1	46.8	56.5	64.8	66.1	60.8	49.7	37.8	29.1
	DBStd	17.38	10.64	9.68	8.01	5.50	5.92	5.58	5.02	4.53	6.29	7.57	7.66	8.83
	HDD50	4025	907	790	679	386	135	7	0	0	5	100	369	647
	HDD65	8055	1372	1210	1144	836	564	262	68	41	152	477	817	1112
	CDD50	1618	0	0	0	0	36	203	457	500	330	90	2	0
	CDD65	173	0	0	0	0	0	8	60	76	27	2	0	0
	CDH74	158	0	0	0	0	0	5	53	92	8	0	0	0
	CDH80	6	0	0	0	0	0	0	1	5	0	0	0	0
(14) Wind	WSAvg	9.2	12.4	11.3	9.5	8.7	6.9	5.9	6.0	6.7	8.0	11.1	12.2	12.2
(15) Precipitation	PrecAvg	32.4	2.1	1.6	1.8	2.6	2.6	3.3	2.8	2.8	3.5	3.3	3.0	2.3
	PrecMax	48.3	4.3	3.0	5.0	5.5	4.9	11.7	4.7	7.2	8.1	7.4	8.1	4.3
	PrecMin	22.7	0.8	0.5	0.2	0.5	0.7	1.5	0.8	0.7	1.1	1.1	0.6	0.7
	PrecStd	5.7	0.8	0.7	1.2	1.3	1.1	2.1	1.3	1.4	1.5	1.5	1.7	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.4	40.4	45.8	52.9	67.4	73.8	78.2	79.6	75.0	68.9	54.1	48.1
		MCWB	39.6	38.1	39.4	43.9	55.9	62.3	69.2	64.8	67.5	62.3	50.7	46.7
	2%	DB	39.2	38.1	41.9	49.0	63.1	70.3	75.6	76.4	73.1	66.5	52.0	44.3
		MCWB	37.2	36.5	38.0	42.6	54.2	59.7	67.7	63.9	67.0	61.1	46.4	42.1
	5%	DB	37.0	36.5	39.5	46.0	59.4	67.6	73.6	74.5	71.2	63.9	50.1	42.2
		MCWB	35.2	34.7	36.5	40.6	53.3	58.8	66.3	64.2	65.8	58.7	45.3	39.8
	10%	DB	34.9	34.5	37.5	44.0	55.2	64.8	71.8	72.8	69.3	60.6	48.0	40.2
		MCWB	33.5	32.6	35.0	39.2	49.7	57.4	65.3	64.3	64.9	55.4	43.6	37.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.4	39.2	40.5	47.2	58.9	64.5	71.6	71.1	68.9	62.4	51.8	47.3
		MCDB	41.1	40.2	41.8	49.8	62.1	68.3	77.1	75.7	72.7	64.9	54.1	48.0
	2%	WB	37.5	36.8	38.3	44.0	55.6	61.2	69.6	69.5	67.6	60.7	49.9	43.2
		MCDB	38.7	38.1	39.8	46.6	60.0	65.2	74.3	73.9	70.9	63.1	52.2	45.0
	5%	WB	35.9	34.7	36.3	41.4	53.2	59.2	68.0	68.3	66.5	58.4	48.0	40.8
		MCDB	37.3	36.0	37.8	44.6	56.7	63.7	72.2	72.5	69.5	60.7	50.1	42.4
	10%	WB	34.2	32.4	34.9	39.6	50.7	57.5	66.4	67.1	65.2	56.2	45.6	38.5
		MCDB	35.7	33.9	36.5	42.8	53.8	61.8	70.9	71.1	68.2	59.1	47.7	40.4
(35) Mean Daily Temperature Range	5% DB	MDBR	11.5	10.9	10.4	9.6	10.9	11.3	9.7	9.5	8.9	8.3	8.0	9.0
		MCDBR	9.5	8.6	10.2	12.6	14.8	13.1	11.0	10.6	9.2	6.6	7.8	8.3
	5% WB	MCWBR	7.7	8.9	8.1	8.3	11.1	9.2	7.0	7.1	6.5	7.3	7.0	8.6
		MCWBR	7.8	9.9	8.5	10.6	15.4	13.0	11.0	10.2	8.7	9.0	8.9	8.9
(40) Clear-Sky Solar Irradiance	taub		0.292	0.298	0.313	0.355	0.373	0.397	0.400	0.389	0.366	0.345	0.311	0.288
		taud	2.440	2.387	2.407	2.346	2.400	2.357	2.376	2.404	2.453	2.476	2.517	2.497
	Ebn at Noon		260	281	292	287	284	276	274	273	270	260	252	251
		Edh at Noon	22	28	32	37	36	38	37	35	30	26	21	19
(44) All-Sky Solar Radiation	RadAvg		381	694	1173	1484	1768	1918	1932	1638	1267	751	401	273
	RadStd		41	68	105	140	165	153	75	87	114	79	56	28

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page